

1 Co-firing copper in a cordierite-type glass-ceramic
2 printed by multi-material vat photopolymerisation: a
3 model-based process design

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5 **Abstract**

Multi-material vat photopolymerisation can print a glass-ceramic body and its copper conductors as one green part, but whether such a part can be fired is not known. Carbon must leave the glass before its pores close, copper must stay metallic, and the two materials must shrink compatibly, all in one temperature range. A verified model couples pyrolysis and steam gasification of char, viscous sintering and crystallisation of the glass, diffusional sintering of copper, and co-firing mechanics, analytically and on the printed voxel grid. Inputs come from the literature, and the glass was calibrated against the published IBM glass-ceramic/copper process, whose atmosphere and burnout temperature the model reproduces. Only steam, with hydrogen held at a fixed fraction of the Cu/Cu₂O equilibrium, removes the char before the glass seals. Fine copper then densifies during the burnout, well before the glass, and the free-strain mismatch reaches 15.6%. A paste and programme optimised together use the retarding effect of char on copper to release both materials at once, cutting the mismatch to 3.7%, but this optimum is fragile. A design that tolerates a factor of two in gasification rate and 15 K in the glass transition holds the mismatch at 9–11% and the copper at 89% IACS in a 16 h cycle.

In three dimensions the optimum reduces the dishing of a printed test vehicle from 622 to 150 μm , and plastic flow of the copper keeps the cooling bow small. Gasification kinetics, glass viscosity, copper sintering viscosity and char yield should be measured first.

6 *Keywords:* Co-firing, Glass-ceramic, Copper metallisation, Vat
7 photopolymerisation, Binder removal, Sintering mismatch

8 **1. Introduction**

9 Packages for millimetre-wave front ends, antenna-in-package modules and
10 wide-bandgap power devices now have to route conductors in three dimensions
11 through a low-loss dielectric, and to do it close to the die [1, 2, 3, 4]. Co-fired
12 ceramics answer part of this demand. Low-temperature co-fired ceramic
13 (LTCC) and glass-ceramic substrates combine low dielectric loss, hermeticity
14 and a thermal expansion close to that of silicon [5, 6]. They are, however,
15 built from stacked tapes: conductors are printed in planes, interconnections
16 run vertically through punched vias, and every change of geometry requires
17 new tooling [7, 5]. The Heterogeneous Integration Roadmap lists additive
18 manufacturing of ceramic packages, including lithography-based ceramic
19 manufacturing, as one way past these limits [8].

20 The conductor of choice in fired packages has long been silver, because it
21 can be co-fired in air. Silver migrates under bias and humidity and diffuses
22 into the glass when fired in oxidising conditions [9, 10, 11]. Copper conducts
23 almost as well, costs less and is far less prone to migration, but it must be
24 fired in an atmosphere that keeps it metallic. In such an atmosphere the
25 organic binder no longer burns off cleanly: part of it is left as char, and char

26 trapped in a glass that has already sealed its pores degrades dielectric loss
27 and can bloat the part [12]. IBM solved this for its glass-ceramic/copper
28 multilayer substrates with a steam-based co-sintering process that gave a
29 dense glass-ceramic with high-conductivity copper [13, 6]. That solution was
30 developed for laminated tapes and its operating window was found empirically.

31 Vat photopolymerisation of ceramic slurries, in particular lithography-
32 based ceramic manufacturing (LCM), produces dense ceramics with features of
33 a few tens of micrometres [14, 15, 16]. It has already been used for monolithic
34 microwave filters and antenna substrates [17, 18, 19], and photocurable LTCC
35 suspensions have been formulated for it [20]. In these parts the metallisation is
36 still added afterwards, by plating or by printing on the fired ceramic [17, 21], or
37 the conductors are jetted into ceramic layers by a separate route [22, 23, 24, 25].
38 Multi-material vat photopolymerisation removes that restriction. Printers
39 such as the CeraFab Multi 2M30 switch between slurries within a layer
40 [26, 27, 28]. Co-sintered ceramic-ceramic parts have been demonstrated [29],
41 as has a zirconia part with titanium-oxide conductors co-fired in a reducing
42 atmosphere [30]. Copper slurries for vat photopolymerisation have been
43 developed separately [31, 32, 33, 34]. In principle, then, a glass-ceramic body
44 and its copper wiring can be printed as one green part. We found no study
45 of whether such a part can be fired.

46 Four requirements make that firing difficult, and they are coupled. First,
47 the char from both slurries must be removed before the glass seals its pores,
48 in an atmosphere that does not oxidise the copper. Second, the two materials
49 must shrink at compatible times and by compatible amounts; otherwise the
50 part cambers, delaminates or cracks around its conductors [35, 36, 37]. Third,

the glass must densify before it crystallises, because crystals stop viscous flow. Fourth, a thermal-expansion mismatch of about 14 ppm K^{-1} between copper and a cordierite-type glass-ceramic must be accommodated on cooling. In tape-based LTCC the second requirement is met by trial, most often by adding ceramic powder to the metal paste [38] or by adjusting the heating rate [39]. Continuum models of co-sintering reproduce camber and mismatch stresses in ceramic and metal–ceramic bilayers [40, 41, 42, 43, 44, 45], but none includes the chemistry of binder removal, and none treats copper with a crystallising glass. The first and third requirements are therefore usually handled separately from the second, although in practice they compete for the same temperature range.

This paper sets out a model-based design of that firing for a copper conductor printed inside a cordierite-type glass-ceramic body. We extend a verified slab model of binder removal and sintering, originally developed for printed copper, to a crystallising glass-ceramic, and couple the two materials through the viscous analogue of laminate theory and a three-dimensional finite-element model built on the printed voxel grid. The model is verified against closed-form solutions and parameterised from published data, with every input tagged by its provenance. With it we (i) map the atmosphere and temperature window in which carbon can be removed from the glass before its pores close while copper stays metallic; (ii) show that fine copper and a fragile glass densify in different temperature ranges and with different curve shapes, and quantify the consequences for camber and conductor damage; (iii) design the copper paste and the firing programme together to bring the two into step, both at nominal inputs and robustly against their uncertainty,

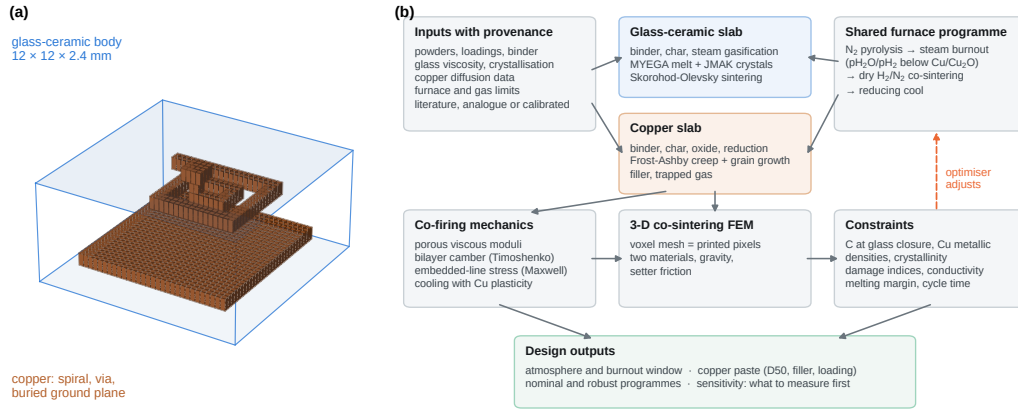


Figure 1: (a) Test vehicle: a $12 \times 12 \times 2.4$ mm glass-ceramic body with a buried copper ground plane, a square spiral and a via to a top pad, printed as one part. (b) Structure of the model. Each material is followed by its own slab model under a shared furnace programme; the two histories are coupled mechanically, analytically and in three dimensions, and the programme and paste are searched against constraints.

and follow the designs through a printed test vehicle in three dimensions; and
 (iv) identify the few material parameters that decide the outcome, and should
 therefore be measured first.

2. Model formulation

2.1. Structure of the model

The co-fired part is represented at two levels (Fig. 1b). At the first level, each material is followed through the firing programme by a one-dimensional slab model that resolves heat transfer, binder decomposition, char formation and removal, pore closure and densification through the thickness of its critical section. The glass-ceramic body and the copper conductor are simulated separately but share the furnace set point and the inlet gas, so both see the

87 same thermal and chemical history. At the second level, the two densification
 88 histories are coupled mechanically: analytically, through the viscous analogue
 89 of laminate theory, and numerically, through a three-dimensional finite-element
 90 model in which both materials occupy the voxels of the printed part. Both
 91 levels use the same constitutive law for the porous body, so a free sample in
 92 the finite-element model reproduces the slab model exactly.

93 The copper slab model was developed and verified for single-material
 94 copper parts printed from a lithography-based slurry; its chemistry and its
 95 calibration against published copper data are summarised in Appendix A. The
 96 additions made here are the crystallising glass-ceramic, the copper paste with
 97 an inert filler, the steam burnout atmosphere, and the co-firing mechanics.

98 *2.2. Binder removal and residual carbon*

99 The photocured binder is lumped into three pseudo-components i (a non-
 100 reactive solvent, a crosslinked network and a char-forming backbone), each
 101 decomposing by first-order kinetics,

$$\frac{dB_i}{dt} = -k_i(T) B_i, \quad k_i(T) = \frac{E_i \beta_{\text{ref}}}{RT_{p,i}^2} \exp \left[-\frac{E_i}{R} \left(\frac{1}{T} - \frac{1}{T_{p,i}} \right) \right], \quad (1)$$

102 where $T_{p,i}$ is the temperature of the corresponding peak in a thermogravimetric
 103 trace at $\beta_{\text{ref}} = 10 \text{ K min}^{-1}$. Writing the rate constant in terms of the peak
 104 temperature keeps the two fitted quantities nearly independent and lets them
 105 be read directly from a single thermogravimetric curve. A fraction χ of the
 106 network and backbone mass remains as char. In a steam-bearing atmosphere

107 the char is gasified by $\text{C} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$ at the rate

$$r_g = a(\varepsilon_o) (1 - f_{\text{cl}}) k_g(T) \frac{c}{M_C} \left(\frac{x_{\text{H}_2\text{O}}}{x_{\text{ref}}} \right)^{1/2} \times \frac{1 + K_H x_{\text{H}_2}^{\text{ref}}}{1 + K_H x_{\text{H}_2}} \left(1 - \frac{Q}{K_{\text{st}}(T)} \right)_+, \quad (2)$$

108 in which c is the char concentration, $a(\varepsilon_o)$ the fraction of pore surface reachable
 109 through open porosity ε_o , f_{cl} the closed-pore fraction, K_H an inhibition
 110 constant for hydrogen adsorption, $Q = x_{\text{CO}}x_{\text{H}_2}/x_{\text{H}_2\text{O}}$ the reaction quotient
 111 and K_{st} the equilibrium constant from NIST-JANAF data [46]. Char in
 112 closed pores is not gasified: once the pores seal, whatever carbon remains
 113 stays in the part. The rate constant k_g is parameterised by a half-life at
 114 800 °C in the reference atmosphere and an activation energy; no measured
 115 value exists for char on either material, so both enter the sensitivity analysis.
 116 A Thiele-modulus estimate ($k_g L^2 / D_{\text{eff}} \sim 10^{-4}$ for $L = 2$ mm) shows that
 117 gasification is kinetically controlled, so steam access is represented by the
 118 open-pore accessibility $a(\varepsilon_o)$ rather than by a diffusion equation.

119 Pyrolysis gas raises the pore pressure. The overpressure is computed
 120 from the compressible Darcy equation with Kozeny–Carman and Knudsen
 121 contributions to the permeability and a polymer-permeation term that carries
 122 the flux before the pore network percolates; it is compared with the green or
 123 brown strength through a safety factor. The same slab tracks the through-
 124 thickness temperature difference and converts it into a thermal-stress index.

125 2.3. Densification

126 Both materials densify according to the Skorohod–Olevsky viscous sinter-
 127 ing model [47]. For a porous body of porosity θ and matrix shear viscosity η ,

128 the stress is

$$\boldsymbol{\sigma} = 2\eta [\varphi(\theta) \dot{\boldsymbol{\epsilon}}' + \psi(\theta) \text{tr}(\dot{\boldsymbol{\epsilon}}) \mathbf{I}] + P_L(\theta) \mathbf{I}, \quad (3)$$

129 with $\varphi = (1 - \theta)^2$, $\psi = \frac{2}{3}(1 - \theta)^3/\theta$ and sintering stress $P_L = 3\gamma(1 - \theta)^2/r$,
 130 where γ is the surface energy and r half the Sauter mean particle diameter.
 131 A free body densifies at $\text{tr}(\dot{\boldsymbol{\epsilon}}) = -(P_L - \Delta p)/(2\eta\psi)$, where Δp is the pressure
 132 of insoluble gas trapped in closed pores. The two materials differ only in η , γ
 133 and r .

134 *Glass-ceramic.* The residual melt is Newtonian. Its viscosity follows the
 135 MYEGA equation [48],

$$\begin{aligned} \log_{10} \eta_g &= \log_{10} \eta_\infty + (12 - \log_{10} \eta_\infty) \frac{T_g}{T} \\ &\times \exp \left[\left(\frac{m}{12 - \log_{10} \eta_\infty} - 1 \right) \left(\frac{T_g}{T} - 1 \right) \right], \end{aligned} \quad (4)$$

136 with $\log_{10} \eta_\infty = -3$ (Pas), glass transition T_g (defined at 10^{12} Pas) and
 137 fragility index m . Crystallisation during firing follows non-isothermal Johnson–
 138 Mehl–Avrami–Kolmogorov kinetics [49] written in terms of the extended-
 139 volume integral,

$$\dot{\xi} = k_c(T) = k_{c,p} \exp \left[-\frac{E_c}{R} \left(\frac{1}{T} - \frac{1}{T_{p,c}} \right) \right], \quad X = 1 - \exp(-\xi^n), \quad (5)$$

140 which avoids the singularity of the rate form at $X = 0$ for $n > 1$. The constant
 141 $k_{c,p}$ is fixed numerically so that the simulated crystallisation exotherm at
 142 10 K min^{-1} peaks exactly at the measured temperature $T_{p,c}$; Kissinger’s closed
 143 form [50] is exact only for $n = 1$ and would misplace the peak by about 1 K
 144 for $n = 2$. Crystals stiffen the melt as rigid inclusions,

$$\eta = \eta_g(T) \left(1 - \frac{X}{X_m} \right)^{-[\eta]X_m} \left(1 - \frac{f_C}{0.64} \right)^{-[\eta]0.64}, \quad (6)$$

the Krieger–Dougherty law [51] with intrinsic viscosity $[\eta] = 2.5$ and maximum packing X_m . The second factor treats residual char particles, volume fraction f_C , the same way. Once X approaches X_m the body stops flowing, which is how crystallisation arrests densification.

Copper. The copper skeleton creeps by diffusion. Its matrix viscosity combines lattice and grain-boundary diffusion [52],

$$\eta_{\text{Cu}} = f_\eta \frac{k_B T G^2}{42 \Omega (D_v + \pi \delta D_b / G)} \left(1 + \frac{C^2}{C_{\text{inh}}^2}\right) \left(1 - \frac{f}{0.64}\right)^{-1.6}, \quad (7)$$

where G is the grain size, Ω the atomic volume and f_η a calibration factor fitted to published copper densification data (Appendix A). The carbon term C (in ppm) describes the retarding effect of residual carbon on copper, and the last factor accounts for an inert ceramic filler of volume fraction f added to the conductor paste, a standard way to slow a metal paste down to the pace of its ceramic [38]. Grain growth is parabolic, driven by grain-boundary diffusion and pinned by porosity. The filler also lowers the conductivity; together with porosity,

$$\frac{\sigma}{\sigma_0} = (1 - \theta)^{3/2} \frac{1 - f}{1 + f/2}, \quad (8)$$

the second factor being the Maxwell Garnett result for insulating inclusions [53].

2.4. Atmosphere and the metallic state of copper

Steam removes char but oxidises copper once the water-to-hydrogen ratio exceeds the equilibrium of $\text{Cu}_2\text{O} + \text{H}_2 \rightleftharpoons 2 \text{Cu} + \text{H}_2\text{O}$,

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} < K_{\text{Cu}}(T) = \exp\left(-\frac{\Delta G^\circ(T)}{RT}\right). \quad (9)$$

164 During the burnout segment the inlet hydrogen is therefore tied to the steam
 165 content, $x_{\text{H}_2} = x_{\text{H}_2\text{O}}/(s K_{\text{Cu}}(T_{\text{B}}))$, with a safety factor $s = 0.2$. The copper
 166 stays metallic by construction, and the steam is as oxidising towards carbon
 167 as the metal allows. This is the principle of the steam co-sintering developed
 168 for glass-ceramic/copper substrates [13, 6]; here it is applied in a model
 169 that says how long, at what temperature and in how much steam. Carbon
 170 gasification itself is thermodynamically unconstrained over the whole window
 171 (Section 4.2); its limits are kinetic.

172 2.5. Mechanics of co-firing

173 *Porous viscosities.* Eq. (3) describes an isotropic, linear viscous solid with
 174 shear viscosity $G_v = \eta\varphi$ and bulk viscosity $K_v = 2\eta\psi$, hence uniaxial viscosity
 175 $E_v = 9K_vG_v/(3K_v+G_v)$, viscous Poisson ratio $\nu_v = (3K_v-2G_v)/[2(3K_v+G_v)]$
 176 and biaxial viscosity $B = E_v/(1 - \nu_v)$. The free linear sintering rate is
 177 $\dot{\epsilon}_{\text{f}} = -P_{\text{L}}/(6\eta\psi)$. All three quantities follow from the slab histories of T , θ ,
 178 G or ξ , and carbon.

179 *Camber.* By the elastic–viscous correspondence principle, a bilayer of two
 180 linear viscous layers bends at a curvature rate given by Timoshenko’s bimetal
 181 solution [54], with elastic moduli replaced by biaxial viscosities and thermal
 182 strain by the difference in free sintering rates [55, 37, 40]:

$$\dot{\kappa} = \frac{6(1+\lambda)^2(\dot{\epsilon}_{\text{f},2} - \dot{\epsilon}_{\text{f},1})}{h[3(1+\lambda)^2 + (1+\lambda\mu)(\lambda^2 + 1/(\lambda\mu))]}, \quad (10)$$

183 with $\lambda = h_1/h_2$, $\mu = B_1/B_2$ and $h = h_1 + h_2$. Layer 1 is copper and layer 2
 184 the glass-ceramic; positive curvature is concave towards the copper. Eq. (10)
 185 is integrated over the firing with the layer thicknesses updated by their

own shrinkage. The reference stack is a 0.1 mm copper plane on 1 mm of glass-ceramic.

Embedded conductors. A conductor line buried in the body cannot curl; its misfit is taken up by stress. We represent line and surrounding body as two Maxwell elements in series, each with a porous elastic modulus and a biaxial viscosity:

$$\dot{\varepsilon}_{f,GC} - \dot{\varepsilon}_{f,Cu} = \dot{\sigma} \left(\frac{1}{E_{Cu}} + \frac{1}{E_{GC}} \right) + \sigma \left(\frac{1}{B_{Cu}} + \frac{1}{B_{GC}} \right). \quad (11)$$

A compliant line follows the body and carries B_{Cu} times the misfit rate; a stiff line forces the body to flow around it. Tension in the copper ($\sigma > 0$) opens voids and breaks the line; compression in the copper means the body is shrinking onto a stiff inclusion and is itself in hoop tension, the classic origin of radial cracks around conductors [35, 36]. Two damage indices compare these stresses with the strength of the weaker partner at that moment,

$$\Pi_{Cu} = \frac{\max(\sigma, 0)}{\sigma_{s,Cu}(\rho_{Cu})}, \quad \Pi_{GC} = \frac{\max(-\sigma, 0)}{\sigma_{s,GC}(\rho_{GC})}, \quad (12)$$

where the strengths σ_s grow from the binder-carried green strength through the neck-controlled brown strength. The series coupling carries an Eshelby factor of order unity [56] that we do not resolve, so the indices are used comparatively. The same coupling is resolved exactly in three dimensions below.

Three-dimensional co-sintering. The printed part is meshed voxel by voxel with trilinear hexahedra, so the mesh is the printed pixel grid. Each element carries its material, porosity and microstructural state. The velocity field satisfies $\nabla \cdot \boldsymbol{\sigma} + \rho_b \mathbf{g} = \mathbf{0}$ with Eq. (3), gravity, and a setter plate beneath the part:

207 zero normal velocity and regularised Coulomb friction ($\mu = 0.6$) iterated to
 208 convergence at each step. Volumetric terms use one-point quadrature, which
 209 avoids locking as $\theta \rightarrow 0$. The problem is assembled with scikit-fem [57] and
 210 solved by conjugate gradients preconditioned with smoothed-aggregation alge-
 211 braic multigrid carrying the six rigid-body modes [58]. Porosity is advanced
 212 with the exact mass-conserving update $1 - \theta^{n+1} = (1 - \theta^n) \exp(-\text{tr } \dot{\epsilon} \Delta t)$.
 213 The temperature and carbon histories are taken from the two slab simulations.
 214 Below 0.3% porosity the constitutive law is evaluated at 0.3%, since the bulk
 215 viscosity otherwise grows without bound and the remaining densification no
 216 longer affects the shape. The simulation ends when the glass-ceramic reaches
 217 98% density or 90% crystallinity, whichever comes first; the remaining linear
 218 shrinkage of the body is then a few tenths of a per cent.

219 *Cooling.* Below a stress-free temperature T_{sf} , where the residual glass stops
 220 relaxing, the thermal-expansion mismatch is carried elastically by the glass-
 221 ceramic and elastoplastically by the copper. The copper layer is treated as
 222 elastic-perfectly plastic with a temperature-dependent yield stress; its mean
 223 stress is capped at yield by a uniform plastic strain, and the curvature follows
 224 from Timoshenko’s elastic solution for the remaining elastic misfit.

225 2.6. Numerical solution, design search and sensitivity

226 The slab equations are integrated with a linearly implicit Rosenbrock
 227 method of orders 2(3) [59] on a geometrically graded mesh that is finest at
 228 the surface. A co-firing evaluation, two slab simulations and the mechanical
 229 post-processing, takes about 6 s. Designs were searched by differential evolu-
 230 tion [60] over eleven variables: copper median particle size, filler fraction and

231 solids loading; burnout temperature, time and steam fraction; temperature
 232 and duration of an intermediate matching hold; heating rate to the peak; peak
 233 temperature and hold. The objective was the largest free-strain mismatch be-
 234 tween the two materials during firing, $\max_t |\varepsilon_{f,GC} - \varepsilon_{f,Cu}|$, which governs both
 235 camber and conductor stress. It was minimised subject to hard constraints
 236 and to a cycle-time budget of 18 h, the ε -constraint form of the trade-off
 237 between cycle time and mismatch. The constraints were: carbon at glass pore
 238 closure below 100 ppm and in the copper below 200 ppm; relative densities
 239 of at least 0.97 for the glass-ceramic and 0.92 for the copper; a crystallised
 240 fraction of at least 0.8; a line damage index below 1; no trapped-gas pressure
 241 above 1.2 times the sintering stress; and a positive margin to the melting
 242 point of copper. Violations entered the objective as a penalty proportional
 243 to their size, so the search is steered back towards feasible designs. The
 244 population of 88 designs was evolved for 15 generations.

245 A second, robust search asked for a design that works across the uncertainty
 246 of the two inputs that decide the timing, the gasification rate and the glass
 247 transition. It evaluates each design at two corners of that uncertainty: slow
 248 gasification ($t_{1/2} = 4$ h) with a low transition ($T_g - 15$ K), where the glass
 249 seals before the char has gone, and fast gasification ($t_{1/2} = 1$ h) with a high
 250 transition ($T_g + 15$ K), where the copper is released before the glass is ready.
 251 The objective is the larger of the two peak mismatches; all constraints must
 252 hold at both corners, and the copper must in addition reach 80% IACS. The
 253 cycle-time budget was relaxed to 20 h, because the first feasible designs found
 254 needed about 20 h; the final design needs 16 h, so the budget did not bind.
 255 The search was seeded with one such feasible design, and a population of 66

256 was evolved for eight generations. The nominal case lies between the corners
257 and was checked on the final design.

258 The influence of uncertain inputs was screened with Morris elementary
259 effects [61]: 16 trajectories through a four-level grid over fourteen factors,
260 240 co-firing evaluations in total, around the optimised design. Twelve
261 factors are the uncertain inputs with the ranges of Table 1; the other two are
262 manufacturing tolerances on the copper paste, -25 to $+33\%$ in particle size
263 and ± 0.05 in filler fraction. For each output the mean absolute elementary
264 effect μ^* measures the overall influence of a factor and the standard deviation
265 σ its non-linearity and interaction with other factors.

266 3. Parameters and calibration

267 3.1. Provenance of the inputs

268 No measurements were made for this study. Every input therefore comes
269 from one of five sources, and Table 1 states which one applies. A value is
270 labelled L when it was reported for the glass-ceramic of the IBM process or
271 for copper itself; A when it was taken from an analogous material, usually a
272 commercial copper-compatible LTCC; C when it was calibrated here against
273 published process observations (Section 3.2); D when it follows from other
274 entries of the table; and E when it is an engineering estimate. Estimated
275 and analogue values that matter enter the sensitivity analysis of Section 4.10,
276 with the ranges given in the table.

277 The reference glass is the MgO-rich cordierite-type glass used by IBM
278 for its copper-compatible substrates (glass #12: 52.5 SiO₂, 22.0 MgO, 22.0
279 Al₂O₃, 1.5 P₂O₅, 0.5 B₂O₃, 1.5 ZrO₂, wt%) [62, 63]. It was chosen because

280 it is the only copper-compatible glass-ceramic for which the firing process,
 281 the atmosphere and the reasons behind the chosen temperatures have been
 282 published in detail. Its viscosity has not been published in a form we could
 283 read, so it was calibrated.

284 *3.2. Calibration of the glass against the IBM process*

285 The patents that describe the IBM steam co-sintering process state, for
 286 glass #12, why the burnout hold sits where it does. The hold is placed
 287 at 785 ± 10 °C, between the annealing and softening points of the glass.
 288 At 750 °C carbon removal would take “prohibitively excessive amounts of
 289 time”; at 830 °C the glass pores close and trap water and binder residue
 290 [62]. A later patent from the same group reports that 33–50% of the total
 291 fired shrinkage occurs during the burnout step [78], and a third places the
 292 onset of crystallisation at about 900 °C, after the burnout hold and before
 293 the crystallisation hold at about 960 °C [79]. We simulated the full worked
 294 schedule of Ref. [62] (N_2 to 200 °C, $\text{H}_2/\text{H}_2\text{O}$ to 450 °C, 2.9 K min^{-1} to the
 295 hold, 6 h at $\text{H}_2/\text{H}_2\text{O} = 10^{-4}$, 0.5 h in N_2 , 2.1 K min^{-1} to 960 °C, 2 h) and
 296 used these statements as targets. The patent specifies the $\text{H}_2/\text{H}_2\text{O}$ ratio but
 297 not the dilution; we assumed 50% steam in nitrogen.

298 Two of them fix the viscosity. Requiring that a 6 h hold at 780 °C
 299 produce 40% of the fired shrinkage with the pores still open, and that the
 300 pores close within 2 h at 830 °C, gives $\log_{10} \eta = 10.62$ at 780 °C and 9.37 at
 301 830 °C. The MYEGA curve through these two points, with $\log_{10} \eta_\infty = -3$,
 302 has $T_g = 733.6$ °C and $m = 32.4$. The remaining observations were not used
 303 in the fit and serve as checks. Between 800 and 860 °C the calibrated viscosity
 304 falls by one decade per 43 K. Giess et al. found that the sintering of an IBM

Table 1: Model inputs. Basis: L, literature value for this material; A, analogue material; C, calibrated in Section 3.2; D, derived; E, estimate. Ranges in the last column are those used in the Morris screening.

Quantity	Symbol	Value	Basis	Source, remark or screening
<i>Glass-ceramic body</i>				
Median particle size	d_{50}	3.0 μm	A	IBM glass powders 2–7 μm
Size span $(d_{90} - d_{10})/d_{50}$	–	1.5	E	
Solids loading of the slurry	ϕ_{GC}	0.45	E	0.40–0.50
Density of the dense body	ρ_s	2600 kg m^{-3}	A	dense α -cordierite glass-cer [64]
Surface energy	γ	0.36 J m^{-2}	L	calculated for a cordierite-ty 0.36, since water vapour low sion of silicate glass [66]
Glass transition (10^{12} Pa s)	T_g	734 $^{\circ}\text{C}$	C	714–754 $^{\circ}\text{C}$
Fragility index	m	32.4	C	28–40
High-temperature viscosity limit	$\log_{10} \eta_{\infty}$	–3	L	[48]
Crystallisation activation energy	E_c	303 kJ mol^{-1}	L	IBM-type high-cordierite 440 kJ mol^{-1} in other MgO– [68, 69]; 250–470 kJ mol^{-1}
Avrami exponent	n	2	E	1.5–3.7; 2.8–3.7 in a related glass [69]
Exotherm peak at 10 K min^{-1}	$T_{\text{p,c}}$	1040 $^{\circ}\text{C}$	C	1000–1080 $^{\circ}\text{C}$
Maximum crystal packing	X_m	0.64	E	random close packing
Density at pore closure	ρ_{cl}	0.92	E	
Thermal conductivity, dense	k	2.8 $\text{W m}^{-1} \text{K}^{-1}$	A	copper co-fired LTCC GL57
Young’s modulus, dense	E_{GC}	128 GPa	A	GL570 [70]
Poisson’s ratio	ν_{GC}	0.25	E	
Thermal expansion coefficient	α_{GC}	3.0 ppm K^{-1}	L	production glass-ceramic su
Flexural strength	$\sigma_{\text{s,GC}}$	210 MPa	L	glass #12 fired in $\text{H}_2/\text{H}_2\text{O}$ [
Stress-free temperature	T_{sf}	690 $^{\circ}\text{C}$	D	strain point ($10^{13.5}$ Pa s) of
<i>Copper conductor</i>				
	16			
Median particle size	d_{50}	1–20 μm	–	design variable; baseline 3 μm
Size span	–	1.2	E	
Solids loading	ϕ_{Cu}	0.40–0.60	–	design variable; baseline 0.5
Inert filler fraction	f	0–0.40	–	design variable; baseline 0
Viscosity factor	f_{η}	0.28	C	Appendix A, from [31]; 0.1

cordierite-type glass over the same range followed the viscosity of the glass, which fell by about one decade per 40 K [65]. The annealing point, taken as 10^{12} Pa s, lies at 734 °C and the dilatometric softening point ($10^{6.6}$ Pa s) at 980 °C, which brackets the 785 °C hold as the patent requires. Stoichiometric cordierite glass is one to four decades more viscous over 780–960 °C [80]. Glass #12 contains more MgO and less Al_2O_3 than cordierite, together with B_2O_3 and P_2O_5 , so a lower viscosity is plausible, but this comparison is not a test of the fit.

The crystallisation kinetics were treated in the same way. With the activation energy measured on an IBM-type glass ($E_c = 303 \text{ kJ mol}^{-1}$ [67]) and $n = 2$, placing the onset of crystallisation (5% crystallised) at 900 °C in the worked schedule requires an exotherm peak at 1040 °C at 10 K min^{-1} . The simulated exotherm then spans 978–1066 °C, a width of about 90 K, which agrees with the 80–100 °C window that the IBM patent associates with the crystallisation exotherm of its glasses [63]. Differential thermal analysis of other MgO– Al_2O_3 – SiO_2 glasses places the α -cordierite peak at 1006–1032 °C [64, 69].

Table 2 summarises the result. The 785 °C hold removes the char and leaves 47% of the shrinkage to the burnout step, inside the 33–50% reported for the process, while the pores stay open. At 830 °C the body densifies completely during the hold and seals its pores. These two rows reproduce the calibration targets and are therefore not independent evidence. The 750 °C row is: it depends on the gasification kinetics, which were not fitted, and it leaves four orders of magnitude more carbon than the 785 °C hold. The model does not reproduce the absolute time scale of the patent’s statement.

Table 2: The IBM worked schedule [62] simulated at three hold temperatures. Shrinkage is the fraction of the total fired shrinkage reached by the end of the 6 h hold; carbon is the residual carbon in the glass-ceramic at the end of the hold.

Hold temperature	750 °C	785 °C	830 °C
Shrinkage during hold	7%	47%	100%
Pores at end of hold	open	open	closed
Carbon at end of hold (ppm)	33	0.006	0.07
Final relative density	0.990	0.990	0.990
Final crystallised fraction	1.00	1.00	1.00
Statement in [62]	too slow	chosen	pores close

330 With the estimated gasification rate, 33 ppm of carbon remain after 6 h at
331 750 °C, which may or may not have been acceptable for IBM; no carbon
332 specification was published. For all three holds the final body is 99% dense
333 and fully crystallised, and its free linear shrinkage is 23.1%.

334 4. Results

335 4.1. Verification

336 The model was checked against closed-form solutions before it was used
337 for design (Appendix B). Isothermal densification of the glass-ceramic follows
338 the Skorohod–Olevsky solution to within 1.3×10^{-4} in porosity for 4 to 16
339 cells (Fig. 2a), and the three-dimensional curvature rate of a free copper/glass-
340 ceramic plate converges to the viscous Timoshenko value at second order,
341 with an error of 0.05% at 0.15 mm voxels (Fig. 2b). The analytical camber of
342 Eq. (10) and the finite-element model therefore describe the same mechanics,
343 and the analytical form can be used inside the design search.

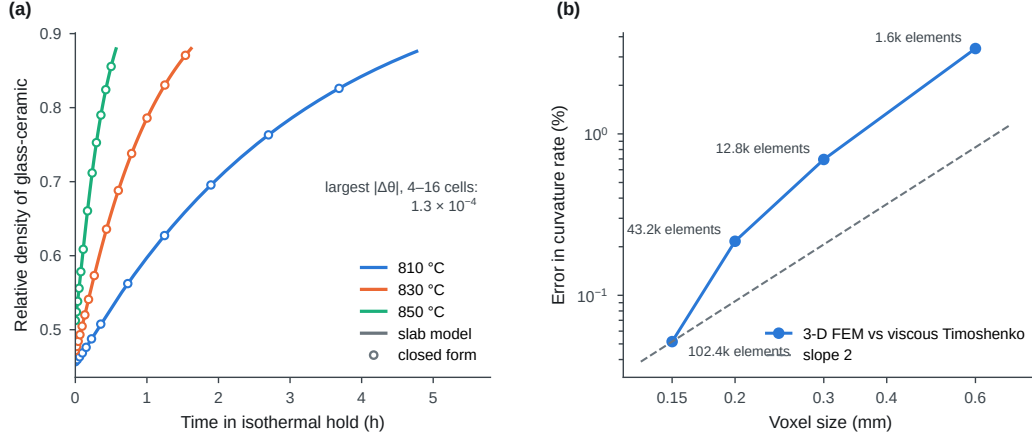


Figure 2: Verification. (a) Isothermal densification of the glass-ceramic slab at 810, 830 and 850 °C (lines, 8 cells) against the closed-form solution integrated along the simulated part temperature (circles); crystallisation and char were switched off. (b) Error of the three-dimensional curvature rate of a free 12×12 mm plate (0.6 mm copper on 1.8 mm glass-ceramic, 900 °C) against the viscous Timoshenko solution, Eq. (10), as the voxel size is refined.

4.2. Atmosphere window for burnout

Fig. 3 places the burnout atmosphere in the plane of temperature and $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$. Copper remains metallic below the orange line, which falls from $10^{6.2}$ at 500 °C to $10^{4.8}$ at 780 °C and $10^{4.1}$ at 1000 °C. Carbon gasification by steam is thermodynamically possible above the dashed line, which lies eight or more decades lower. The two constraints therefore leave a wide band of permissible atmospheres, and the choice within it is kinetic. At 780 °C, 30% steam with hydrogen at 20% of the copper limit ($p_{\text{H}_2\text{O}}/p_{\text{H}_2} = 1.2 \times 10^4$) gives a char half-life of 0.43 h. Forming gas (4% H_2) humidified through a bubbler at 20 °C, a common choice for copper, contains 2.3% water; its half-life at the same temperature is 3.1 h, seven times longer, because the steam partial

355 pressure is low and hydrogen inhibits the reaction.

356 The operating point of the IBM process, $H_2/H_2O = 10^{-4}$ at 785 ± 10 °C
357 [62], lies at 17% of the copper limit and almost coincides with the steam
358 point in Fig. 3. The rule used here, hydrogen tied to steam so that the
359 water-to-hydrogen ratio sits at a fixed fraction of the copper limit, therefore
360 reproduces the atmosphere that was found empirically for tape-cast substrates.
361 It also says how the ratio should change if the burnout temperature changes:
362 by a factor of about 2.5 per 100 K, following $K_{Cu}(T)$.

363 *4.3. Removing carbon before the glass seals*

364 The char has to leave the glass before its pores close, and the glass
365 carries far more of it than the copper does. After pyrolysis in nitrogen the
366 glass-ceramic contains 22 500 ppm carbon relative to its solids, the copper
367 5 300 ppm. The difference follows from the densities: both slurries hold a
368 similar volume of binder, but a unit volume of glass-ceramic solids weighs
369 a third as much as the same volume of copper. Carbon on the glass is also
370 the more harmful, since whatever is sealed in reacts with the water released
371 during crystallisation and bloats the part [62].

372 Fig. 4 maps the carbon left in the glass-ceramic at the moment its pores
373 close, as a function of burnout temperature and hold time. In steam (Fig. 4a)
374 the 100 ppm contour runs from 16 h at 710 °C to 1.1 h at 830 °C, and the
375 window closes abruptly at 850 °C, where the glass seals during the hold. The
376 IBM hold of 6 h at 785 °C falls in the region with less than 10 ppm. In
377 humidified forming gas (Fig. 4b) only the longest hold, 16 h at 790 °C, brings
378 the carbon down to 350 ppm; every other combination leaves more than
379 1000 ppm. Residual carbon also slows the densification of the glass. Acting

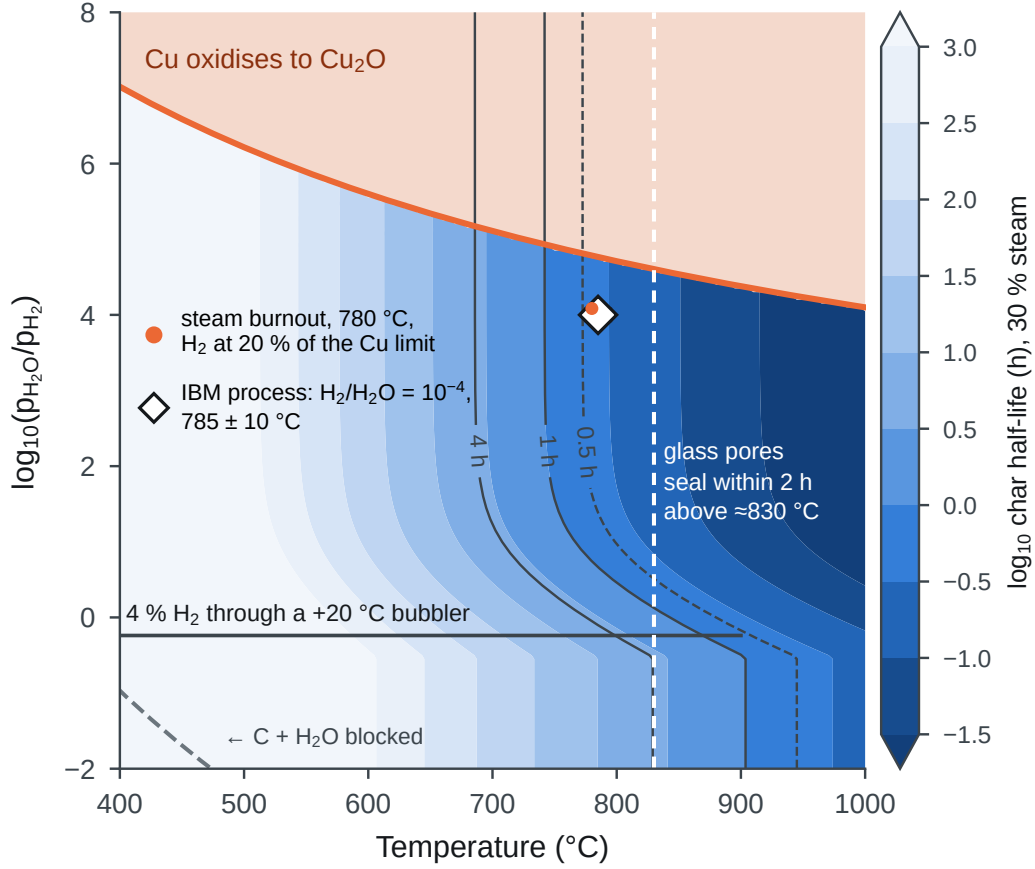


Figure 3: Burnout atmosphere window. Colour: half-life of char in 30% steam at the given $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$ (reference kinetics of Table 1). Copper oxidises above the orange line; carbon cannot be gasified to the left of the dashed line ($p_{\text{CO}} = 10^{-4}$ atm). Circle: steam burnout used in this work. Diamond: operating point of the IBM process [62]. The white line marks the temperature above which a 2 h hold seals the pores of the calibrated glass.

380 as rigid inclusions, the 0.3–2% of carbon retained in forming gas lowers the
 381 final density from 0.99 to 0.96–0.99. Steam is therefore not an option for this
 382 material pair but a requirement.

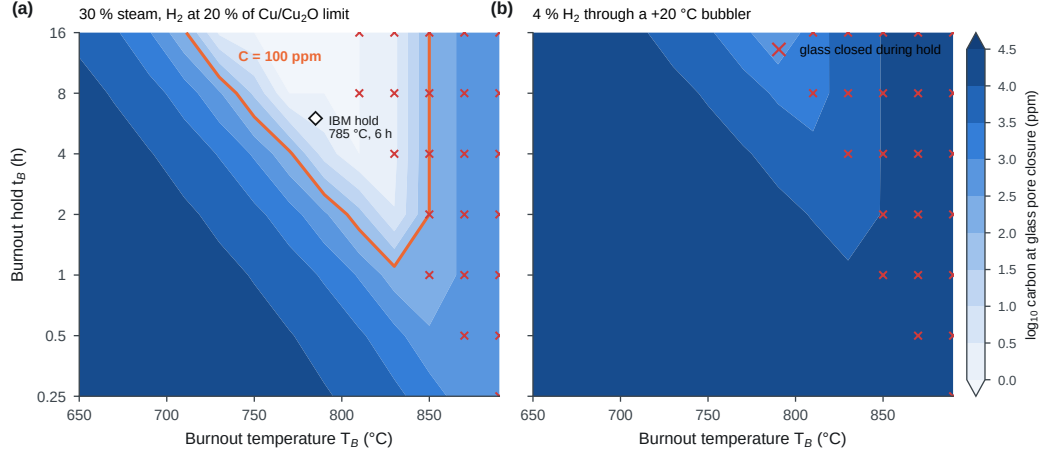


Figure 4: Carbon remaining in the glass-ceramic when its pores close, for burnout holds of temperature T_B and duration t_B within the baseline programme. (a) 30% steam with hydrogen at 20% of the Cu/Cu₂O limit; (b) 4% H₂ humidified at 20 °C. Orange: 100 ppm contour. Crosses: the glass sealed during the hold. Diamond: the IBM burnout hold [62].

4.4. Two densification ranges

Fig. 5 compares the free sintering of the two materials after binder removal at 5 K min⁻¹ in dry hydrogen. The glass-ceramic shrinks by 23% between 850 and 933 °C (5 and 95% of its shrinkage) and then stops, because it crystallises: 93% of the glass has crystallised at 1060 °C. Fine copper densifies much earlier, and over a much wider range. A 3 μ m powder shrinks between 454 and 702 °C; a 1 μ m powder between 323 and 463 °C. Only for powders of 12 μ m or coarser does the midpoint of densification fall inside the range of the glass, and these powders do not reach high density before 1060 °C: 83% for 12 μ m and 60% for 20 μ m.

The two curves also differ in shape. The glass densifies over a range of about 80 K, as a Newtonian melt whose viscosity falls steeply with tempera-

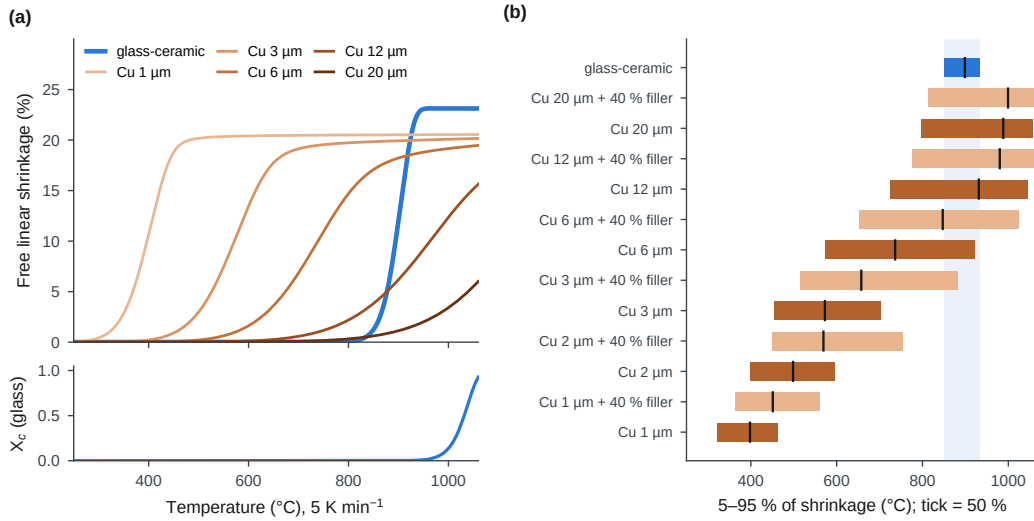


Figure 5: Free sintering of the two materials at 5 K min⁻¹ in dry 4% H₂ after binder removal. (a) Linear shrinkage of the glass-ceramic and of copper of different median particle size, and crystallised fraction of the glass. (b) Temperature range between 5 and 95% of the final shrinkage, with the midpoint marked, for copper with and without 40 vol% inert filler; the shaded band is the range of the glass-ceramic. Bars that reach 1060 °C had not finished densifying.

395 ture. Copper densifies over 140–350 K, because diffusional creep has a lower
 396 activation energy and grain growth slows it as it proceeds. Adding an inert
 397 filler to the paste delays and stretches the copper curve (Fig. 5b): 40% filler
 398 shifts the midpoint of 3 μm copper from 573 to 657 °C and widens the 5–95%
 399 range from 248 to 367 K. Neither particle size nor filler, on its own, gives
 400 copper a shrinkage curve that is as late and as steep as that of the glass. The
 401 matching has to come partly from the furnace programme.

402 4.5. *Co-firing with an unmatched paste*

403 The baseline design combines a 3 μm copper paste without filler with a
404 programme that follows the IBM pattern: pyrolysis in nitrogen to 460 °C, a
405 4 h burnout at 780 °C in 30% steam, and sintering at 960 °C for 1 h in dry 4%
406 H_2 (Fig. 6a). Chemically, the design works. The glass-ceramic retains 33 ppm
407 of carbon when its pores close and reaches 99.1% density and full crystallinity;
408 the copper is 97.6% dense, retains 81 ppm of carbon and conducts at 96%
409 IACS.

410 Mechanically, it fails. As soon as steam begins to remove the char, the
411 copper starts to sinter, while the glass is still rigid. At the end of the burnout
412 hold the logarithmic shrinkage of the copper is 21% and that of the glass 7%;
413 the free-strain mismatch peaks at 15.6%, and a free bilayer of 0.1 mm copper
414 on 1 mm glass-ceramic would reach a curvature of 58 m^{-1} , concave towards
415 the copper. The damage index of an embedded line reaches $\Pi_{\text{Cu}} = 1.6$ at
416 680 °C, during the ramp to the burnout temperature, when the line is pulled
417 in tension by the rigid body around it while its own necks are still weak. The
418 glass then catches up. Its shrinkage overtakes that of the copper at 889 °C,
419 the bilayer bends the other way and reaches -259 m^{-1} (a radius of 3.9 mm)
420 at 930 °C, and the final shrinkages differ by 4%: 26.3% for the glass-ceramic
421 against 22.3% for the copper, both as logarithmic strains. For this material
422 pair the difficulty lies less in the final shrinkage than in the order in which
423 the two materials shrink.

424 4.6. *Copper paste design maps*

425 Fig. 7 shows how far the materials alone can repair the baseline, with the
426 programme left unchanged. The upper row varies the copper paste. Coarser

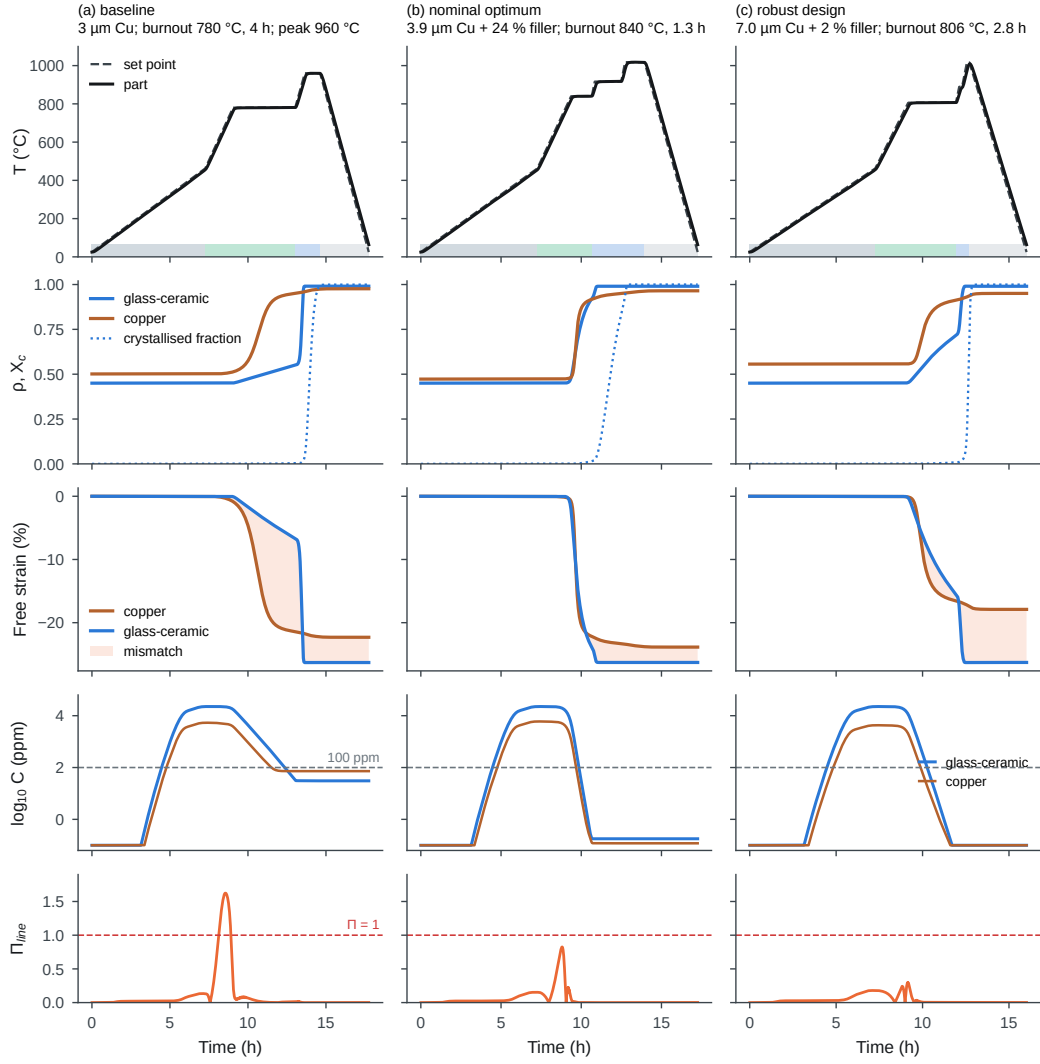


Figure 6: Co-firing histories of (a) the baseline, (b) the nominal optimum and (c) the robust design of Table 3. From top: furnace set point and part temperature, with the atmosphere of each segment shown by the band at the bottom (grey, N_2 ; green, steam; blue, dry H_2/N_2); relative densities and crystallised fraction of the glass; free logarithmic shrinkage of each material, the shaded area being the mismatch; carbon in the glass-ceramic and damage index of an embedded copper line.

427 powder and filler both delay the copper, and the damage index of an embedded
 428 line falls below 1 for particle sizes of $4.5\ \mu\text{m}$ and above (Fig. 7b). The peak
 429 mismatch falls more slowly. Its lowest values, about 9.5%, are reached at
 430 $6\ \mu\text{m}$ with 30% filler and at $9\ \mu\text{m}$ without filler; coarser pastes are released
 431 too late and the mismatch rises again, now with the copper behind the glass.
 432 Every step towards a later copper costs density or conductivity (Fig. 7c,d).
 433 Filler lowers the conductivity directly, by 14 points of IACS for every 10 vol%,
 434 while coarse powder lowers it through porosity: $18\ \mu\text{m}$ copper reaches only
 435 76% density. Among the pastes that meet the density limit, $9\ \mu\text{m}$ copper
 436 without filler gives the best compromise: 9.5% peak mismatch, a damage
 437 index of 0.38 and 91% IACS. None of them reduces the final camber, which
 438 stays between -190 and $-260\ \text{m}^{-1}$ for every paste that meets the density
 439 limit. The final camber is set by the difference in final shrinkage, which the
 440 paste barely changes.

441 The lower row varies the glass, with the baseline paste. The glass transition
 442 decides whether carbon is trapped. Below about $695\ ^\circ\text{C}$ the glass seals during
 443 the $780\ ^\circ\text{C}$ burnout and retains about 3000 ppm (Fig. 7e). Above that, the
 444 carbon stays below 100 ppm, but the higher the transition, the later the glass
 445 densifies, and the crystallisation peak has to rise with it if the glass is to reach
 446 full density before it crystallises (Fig. 7f). The crystallisation peak must not
 447 be too high either: above about $1080\ ^\circ\text{C}$ the glass is less than 80% crystalline
 448 after 1 h at $960\ ^\circ\text{C}$ (Fig. 7g). The most interesting region is the narrow band
 449 around $T_g = 700\ ^\circ\text{C}$ (Fig. 7h). There the glass densifies during the burnout
 450 hold, together with the copper, and the peak mismatch falls to 6.5% while
 451 the carbon at closure stays near 50 ppm. A glass chosen to match the copper

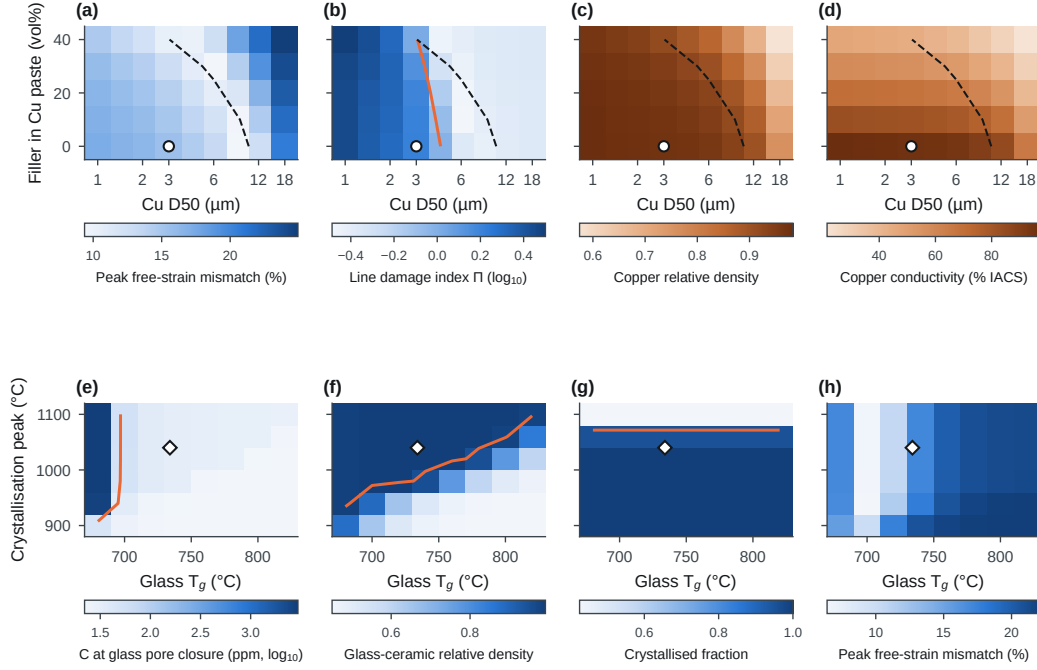


Figure 7: Design maps at the baseline programme. Top: copper paste, median particle size against filler fraction, with the baseline paste marked (circle): (a) peak free-strain mismatch; (b) damage index of an embedded line, orange contour $\Pi = 1$; (c) copper density; (d) conductivity. The dashed line is the density limit $\rho_{\text{Cu}} = 0.92$; pastes to its right fail it. Bottom: glass, transition temperature against crystallisation peak, with the calibrated glass marked (diamond) and the baseline paste: (e) carbon at pore closure, orange contour 100 ppm; (f) density of the glass-ceramic, orange contour 0.97; (g) crystallised fraction, orange contour 0.8; (h) peak mismatch.

452 would thus do more than any paste, but only within a window of about 20 K
 453 in T_g , and with little margin on carbon at its lower edge.

454 4.7. A matched paste and programme

455 Neither the paste nor the glass alone brings the peak mismatch below about
 456 6%, so the paste and the programme were searched together (Section 2.6).

457 The optimum found within an 18 h cycle is listed in Table 3 and its history
458 is shown in Fig. 6b. It uses a 3.9 μm copper powder with 24 vol% filler.
459 The burnout is short and hot: 1.3 h at 840 °C in 81% steam, just below the
460 temperature at which the glass would seal during the hold (Fig. 4a). A 1.7 h
461 hold at 916 °C follows, and then 1.2 h at 1018 °C. The peak mismatch falls
462 from 15.6 to 3.7%, the damage index of an embedded line from 1.6 to 0.83,
463 and the final camber of the reference bilayer from -254 to -76 m^{-1} . The
464 glass-ceramic is 99.0% dense and fully crystallised, with 0.2 ppm of carbon at
465 pore closure.

466 The optimiser found a mechanism rather than a compromise. The copper
467 cannot sinter while it carries char (Section 4.5), so the burnout releases it.
468 Placing the burnout just below the coalescence of the glass, and making it
469 fast with a high steam fraction, releases the copper at the moment the glass
470 begins to flow. The two materials then densify together (Fig. 6b), and the
471 filler slows the copper just enough to follow the glass rather than lead it. The
472 hold at 916 °C lets the glass finish densifying before it crystallises, and the
473 peak at 1018 °C finishes the copper.

474 The optimum has two weaknesses. The first is conductivity: 24% filler
475 limits the copper to 64% IACS. The second is robustness. The coincidence on
476 which the optimum rests is sharp, and Table 3 shows what happens when the
477 two least certain inputs are varied within plausible bounds. If the char gasifies
478 half as fast as assumed, the carbon at closure rises from 0.2 to 76 ppm and the
479 peak mismatch from 3.7 to 11.1%; at a quarter of the assumed rate the carbon
480 limit is exceeded by an order of magnitude. A glass transition 15 K higher or
481 lower than calibrated raises the peak mismatch to about 10%. A factor of

two in the viscosity of copper matters less (5.8–6.2%). A design that relies on the timing of char removal is therefore only as good as the gasification kinetics and the viscosity of the glass, and neither has been measured for this material pair.

The robust search (Section 2.6) therefore asked for a design that meets every constraint at two corners of this uncertainty and, in addition, keeps the copper above 80% IACS. The result, listed in the last column of Table 3 and shown in Fig. 6c, follows a different strategy. It uses a coarser $7\text{ }\mu\text{m}$ copper with almost no filler and a higher solids loading, and burns out for 2.8 h at $806\text{ }^{\circ}\text{C}$ in 83% steam, some 40 K below the temperature at which the glass would seal during the hold. It then heats quickly to a short peak at $1017\text{ }^{\circ}\text{C}$ and needs 16.0 h in all. Both materials start to densify during the burnout hold, the copper once its char has gone and the glass because it is already soft. Being coarse, the copper runs only a few per cent ahead of the glass; the glass overtakes it during the heating to the peak. At nominal inputs the peak mismatch is 9.3%, the conductivity 89% IACS and the damage index 0.30. In both corners the carbon stays below 50 ppm, the damage index below 0.43 and the peak mismatch at or below 11.4%. The single-factor changes of Table 3 move its peak mismatch only between 9.3 and 9.9%. Robustness has a price: the nominal mismatch is two and a half times that of the sharp optimum, and the final camber, -187 m^{-1} , is larger. Most of the remaining mismatch is a difference in final shrinkage, 8.4%, because the paste now carries more solids than the glass slurry (56 against 45 vol%), which helps the copper reach density without filler. A gasification rate four times slower than assumed defeats both designs, which leave 940 and 1295 ppm of carbon

Table 3: Baseline and optimised designs. Robustness: the nominal optimum and the robust design re-evaluated with the char gasification half-life ($t_{1/2}$, nominal 2 h) or the glass transition (T_g , nominal 734 °C) changed as indicated. Values that violate a constraint are in bold.

	Baseline	Nominal optimum	Robust design
<i>Paste and programme</i>			
Copper d_{50} (μm) / filler (vol%) / solids (vol%)	3.0 / 0 / 50	3.9 / 24 / 47	7.0 / 2 / 56
Burnout: T (°C) / time (h) / steam (%)	780 / 4.0 / 30	840 / 1.3 / 81	806 / 2.8 / 83
Matching hold: T (°C) / time (h)	–	916 / 1.7	889 / 0.2
Peak: T (°C) / time (h)	960 / 1.0	1018 / 1.2	1017 / 0.2
Cycle time (h)	17.7	17.3	16.0
<i>Outcome at nominal inputs</i>			
Carbon at glass pore closure (ppm)	33	0.2	0.03
Glass-ceramic density / crystallised fraction	0.991 / 1.00	0.990 / 1.00	0.990 / 1.00
Copper density / conductivity (% IACS)	0.976 / 96	0.965 / 64	0.950 / 89
Peak free-strain mismatch (%)	15.6	3.7	9.3
Line damage index Π_{Cu}	1.63	0.83	0.30
Final camber of reference bilayer (m^{-1})	–254	–76	–187
<i>Robustness: carbon at closure (ppm) / peak mismatch (%)</i>			
$t_{1/2} = 4$ h	–	76 / 11.1	37 / 9.9
$t_{1/2} = 8$ h	–	1295 / 17.9	940 / 14.4
$T_g - 15$ K	–	84 / 9.9	0.06 / 9.4
$T_g + 15$ K	–	0.06 / 9.5	0.03 / 9.8

507 in the glass.

508 4.8. The printed test vehicle

509 The three designs were then fired in three dimensions as the printed test
510 vehicle of Fig. 1a: a $12 \times 12 \times 2.4$ mm glass-ceramic block with a buried

511 copper ground plane, a square spiral and a via to a top pad, meshed with
512 12 800 voxels of 0.3 mm and resting on a setter. Each run starts after binder
513 removal and ends when the glass-ceramic is 98% dense, at 903–917 °C (Fig. 8).
514 The setter keeps the lower face flat. The distortion therefore appears on the
515 upper face, which dishes over the buried plane.

516 The buried plane dominates the result. In the baseline the copper has
517 already densified when the glass begins to flow, so the plane acts as a rigid
518 inclusion. The block shrinks by only 14.9% in plane, against 23.1% for the
519 free glass-ceramic, and the glass above the plane, unable to shrink laterally,
520 densifies by shrinking more in thickness. The upper face ends 622 μm lower
521 at the centre than at the rim (Fig. 8c), about a third of the fired thickness. In
522 the nominal optimum the copper and the glass densify together. The plane
523 then offers little resistance, the in-plane shrinkage rises to 21.2%, and the
524 relief of the upper face falls to 150 μm . The robust design lies between them:
525 18.3% in-plane shrinkage and 317 μm of relief, consistent with its larger final
526 mismatch. The via and pad raise a small plateau at the centre of every part.
527 The ranking of the three designs by relief follows the ranking by mismatch
528 of the one-dimensional analysis, which supports the use of that analysis for
529 design. The absolute values, however, depend on the area of copper. A plane
530 that covers 56% of the part, as here, is the most demanding case, and a real
531 layout would divide it or add dummy metal in the rim.

532 4.9. *Cooling from the stress-free temperature*

533 Below about 690 °C, the strain point of the calibrated glass, mismatch
534 stresses no longer relax, and the expansion mismatch between copper (16.5–
535 22 ppm K⁻¹ over this range) and the glass-ceramic (3 ppm K⁻¹) is carried

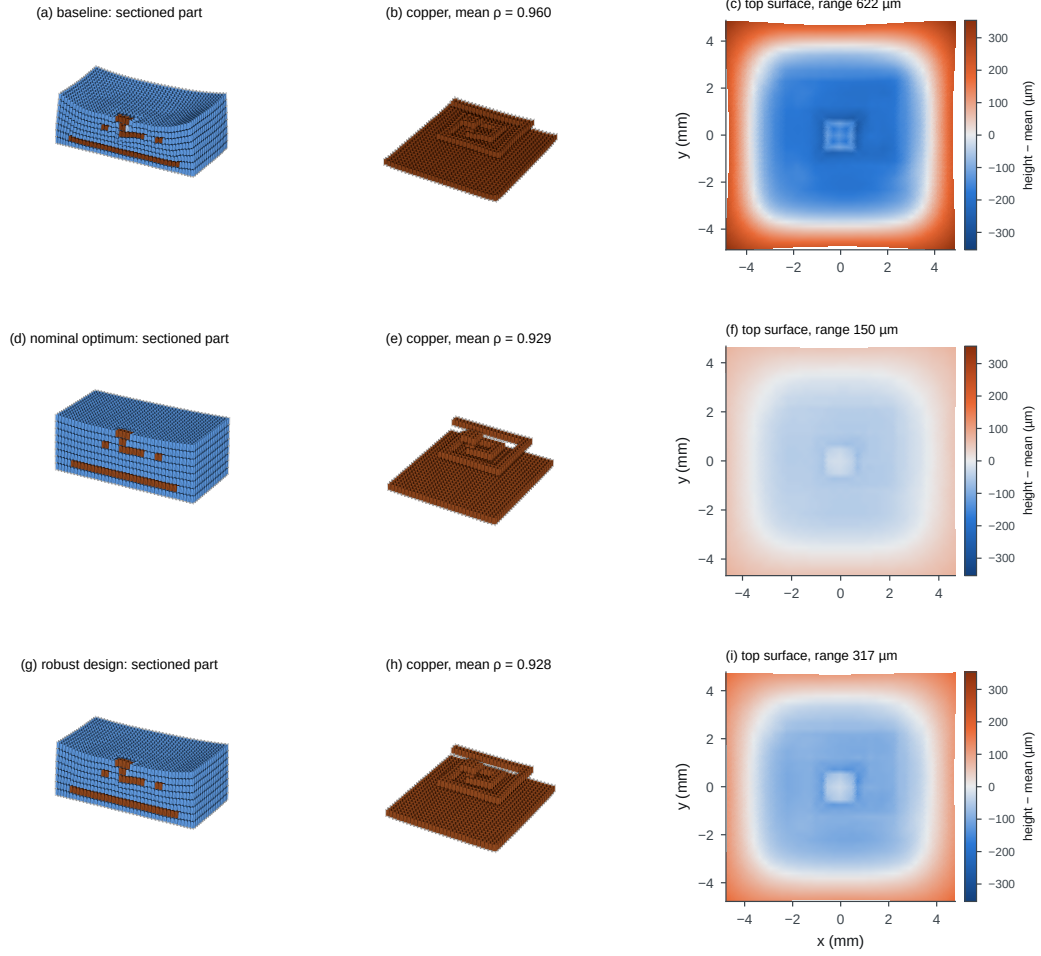


Figure 8: Three-dimensional co-sintering of the printed test vehicle (0.3 mm voxels, setter support, gravity) for the three designs of Table 3, from binder removal to 98% density of the glass-ceramic. Left: part sectioned through its centre, showing the buried plane, the spiral and the via. Centre: the copper, coloured by density. Right: height of the upper face relative to its mean; the lower face is held flat by the setter.

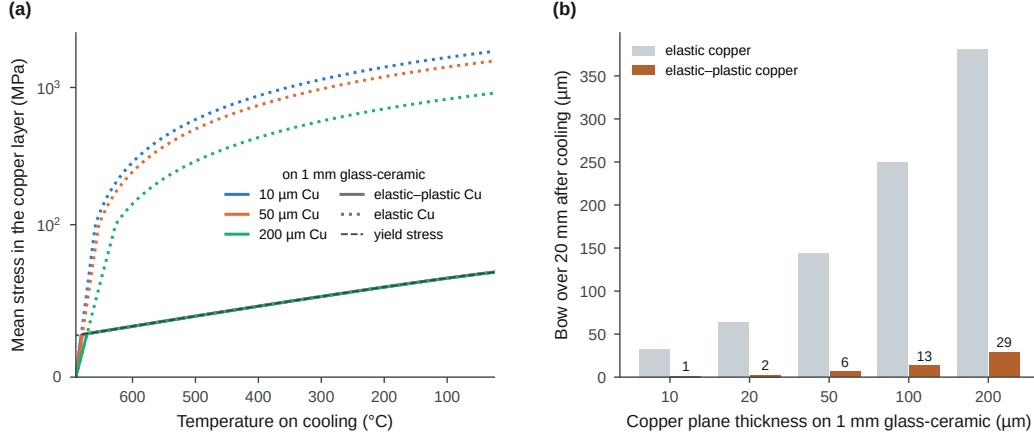


Figure 9: Cooling of a copper plane on 1 mm of glass-ceramic from the stress-free temperature (690 °C). (a) Mean stress in the copper layer for elastic and elastic-perfectly plastic copper; the plastic curves coincide with the temperature-dependent yield stress. (b) Bow over a 20 mm span at room temperature.

536 elastically. Fig. 9 shows the consequences for a copper plane on 1 mm of glass-
537 ceramic. If copper remained elastic, its mean stress would exceed 900 MPa
538 at room temperature and a 20 mm stack with 100 μm of copper would bow
539 by 250 μm . Annealed copper cannot carry that stress. It yields within
540 the first 10 K of cooling and then follows its yield stress, which rises from
541 about 25 MPa at 690 °C to 69 MPa at room temperature. The bow falls
542 accordingly, to 13 μm for 100 μm of copper and to 6 μm for 50 μm , and
543 the glass-ceramic stays in mild compression (-7 MPa for 100 μm of copper).
544 Plastic accommodation makes the thermal-expansion mismatch a secondary
545 problem compared with the sintering mismatch, provided the copper is dense
546 and ductile. The price is cyclic plasticity in the conductor, which matters for
547 thermal cycling in service but is outside the scope of this study.

548 4.10. Which inputs decide the outcome

549 Fig. 10 ranks the fourteen uncertain inputs by their elementary effects
550 around the nominal optimum. Elementary effects are expressed per full range
551 of each factor. One input dominates. Across its range of 0.5–8 h, the half-life
552 of char gasification changes the peak mismatch by 16 percentage points on
553 average and the carbon at glass closure by several orders of magnitude, from
554 well below 1 ppm to above the 100 ppm limit. It also comes second for
555 copper density, because char that is not removed keeps the copper from
556 sintering. The carbon trapped in the glass depends next on the fragility and
557 transition temperature of the glass, an order of magnitude less strongly. The
558 density of the glass-ceramic is governed by its own properties: the transition
559 temperature, the particle size and the crystallisation kinetics. The density
560 and conductivity of the copper depend on the viscosity factor f_η and on the
561 filler. The peak mismatch, the quantity that the design sought to minimise,
562 responds to almost every input after the gasification rate: f_η , the glass
563 transition, the char yield, and the particle size and solids loading of the glass.
564 For every output that matters, σ is of the same order as μ^* . The effects are
565 therefore strongly non-linear and interact, which is what a design based on
566 the coincidence of two processes should be expected to show. The tolerances
567 on the copper paste itself (particle size $\pm 30\%$, filler $\pm 5 \text{ vol}\%$) matter mainly
568 for the damage index of a buried line, together with f_η and the char yield.
569 The Avrami exponent and the activation energy of crystallisation matter only
570 for the density of the glass-ceramic, and the crystallised fraction (not shown)
571 remained complete in every run.

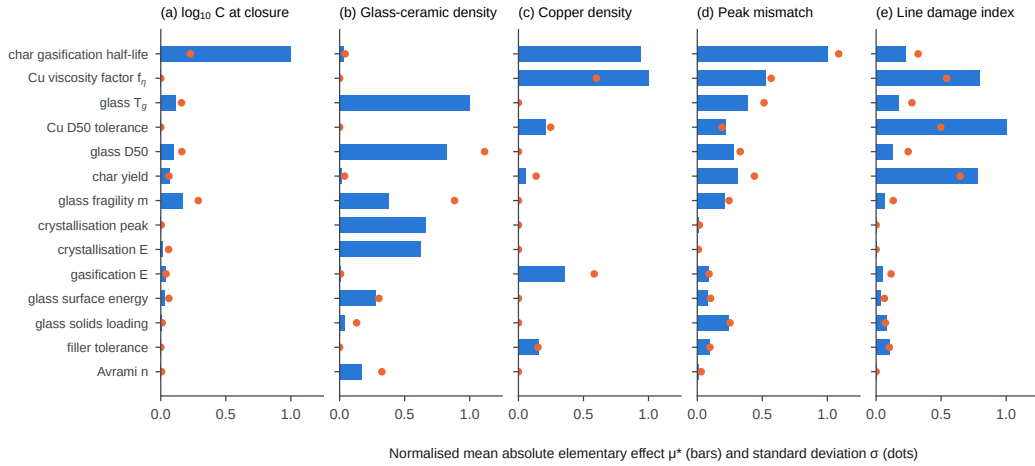


Figure 10: Morris screening of the fourteen uncertain inputs around the nominal optimum (16 trajectories, 240 co-firing evaluations). Bars: mean absolute elementary effect μ^* ; dots: standard deviation σ of the elementary effects, both normalised by the largest μ^* of each output. Factors are ordered by their summed normalised importance. Ranges as in Table 1; the copper particle size and filler fraction are varied as manufacturing tolerances around the design.

5. Discussion

5.1. The order of events in co-firing

Taken together, the results fix an order of events that any firing programme for this material pair has to respect. Pyrolysis in nitrogen leaves about 2% of char, relative to the solids, in the glass-ceramic and about 0.5% in the copper. The char must then be gasified before the glass seals its pores. With the atmosphere held at a fixed fraction of the copper–oxide equilibrium, this requires hours at 770–800 °C and is the slowest step of the process. Removing the char has a second effect, which is easy to overlook: it releases the copper. Residual carbon retards the sintering of copper, so a fine powder that has

582 been held back by its own char starts to densify as soon as the steam clears it,
583 at 680–780 °C in the baseline and well before the glass. In silver-metallised
584 LTCC fired in air this coupling does not arise, because the binder burns off
585 completely at low temperature and the metal is free to sinter from then on.
586 Here the burnout step is also, whether intended or not, the first sintering step
587 of the conductor. The glass then densifies over a range of about 80 K and
588 stops when it crystallises, and below the strain point of the residual glass the
589 expansion mismatch is taken up by plastic flow of the copper.

590 This order explains why neither lever works alone. The furnace programme
591 cannot delay the densification of fine copper beyond the burnout, because the
592 burnout has to happen below the coalescence of the glass. The paste cannot
593 be made as late and as steep as the glass (Fig. 5) without losing conductor
594 density. The two designs of Table 3 are the two ways out. The nominal
595 optimum uses the coupling between char and copper to start both materials
596 together. It reaches the lowest mismatch but depends on the timing of char
597 removal, which is the least known input of the model. The robust design gives
598 up part of that gain. It relies on a coarse copper that is intrinsically slow,
599 and it tolerates a factor of two in the gasification rate together with 15 K
600 in the glass transition. Which of the two is preferable depends on what has
601 been measured. Once the gasification kinetics and the viscosity of the glass
602 are known to within these bounds, a design closer to the nominal optimum
603 becomes safe.

604 5.2. *Relation to tape-based co-firing*

605 The rule used for the atmosphere, hydrogen tied to steam so that the
606 water-to-hydrogen ratio sits at a fixed fraction of the copper limit, reproduces

607 the operating point of the IBM process (Fig. 3), and the calibrated glass
608 reproduces the reasons given for its burnout temperature (Table 2). The
609 model does not add a new principle to that process. What it adds is a
610 quantitative account: how long the burnout must last at a given temperature
611 and steam content, how much carbon is left when the pores close, and how
612 the choice of copper powder interacts with the programme. In silver-based
613 LTCC, camber is controlled empirically by adjusting the paste [38] or the
614 heating rate [39]. For copper in a crystallising glass the ladder of Fig. 5 shows
615 why paste formulation alone is not sufficient. The heating programme sets
616 the order in which the materials are released to sinter, and the paste only
617 fine-tunes the rate once they are.

618 Printing changes the problem in three respects. First, a photocurable
619 slurry carries more binder than a cast tape: at 45 vol% solids the glass-ceramic
620 shrinks by 23% linearly, against 12.7–15.8% in the plane of commercial LTCC
621 tapes [81, 82], and it holds correspondingly more char. Second, conductors
622 are no longer confined to printed planes connected by punched vias. The test
623 vehicle contains a spiral, a buried plane and a via defined in one build, and
624 the three-dimensional model shows that their densification and distortion
625 follow from the same constitutive law as in a bilayer. It also shows that a
626 large buried plane, which tape technology would mesh or perforate, dominates
627 the distortion; a printed layout can divide such a plane freely. Third, printed
628 ceramics shrink anisotropically between the build direction and the plane, as
629 laminated tapes do: unconstrained IBM laminates fired in $\text{H}_2/\text{H}_2\text{O}$ shrink
630 about half as much through the thickness as in the plane [79]. The model
631 treats both materials as isotropic. Anisotropy would change the camber and

632 the shape of the fired part, but not the order of events described above.

633 5.3. *Limitations*

634 The results rest on published data and a model; no part has been fired.
635 Three inputs deserve particular caution. The viscosity of the glass was
636 calibrated against statements in patents, not measured, and it describes
637 one glass composition; a different cordierite-type glass will shift the whole
638 window. The gasification kinetics of photopolymer char on either material
639 are unknown and were estimated, so the absolute times of the burnout carry
640 the largest uncertainty of the study (Section 4.10). The viscosity factor of
641 copper was fitted to a coarser powder than those considered here. In addition,
642 the model omits several phenomena that can matter in practice: dissolution
643 of copper oxide in the glass and interdiffusion at the interface, adhesion and
644 delamination, the water released by the glass during crystallisation, the effect
645 of residual carbon on dielectric loss, and cyclic plasticity of the copper in
646 service. The damage indices compare stresses with strength estimates and are
647 meant for ranking designs, not for predicting failure. The three-dimensional
648 runs stop at 98% density of the glass-ceramic, so the final densification of the
649 copper at the peak temperature, inside an already rigid body, is represented
650 only in the slab model.

651 5.4. *What to measure first*

652 The sensitivity analysis (Fig. 10) turns the limitations into a short list of
653 measurements, ordered by their value to the design:

- 654 1. The gasification kinetics of the photopolymer char on each material, in
655 the steam–hydrogen atmosphere of the burnout. Interrupted burnout runs

656 followed by carbon analysis at three temperatures would fix both the rate
657 and its activation energy. No other quantity has as large an effect on the
658 carbon trapped in the glass and on the mismatch.

- 659 2. The viscosity of the glass between 750 and 900 °C, which fixes T_g and
660 m . Beam bending or parallel-plate viscometry of the glass would give it
661 directly, and so would isothermal dilatometry of the printed glass-ceramic,
662 interpreted with the closed form of Appendix B.
- 663 3. The sintering viscosity of the copper slurry, f_η , from dilatometry of printed
664 copper coupons in dry hydrogen after binder removal.
- 665 4. The char yield of both cured slurries, from thermogravimetry in nitrogen.

666 The particle size and solids loading of the glass also matter, but they are
667 specified by the supplier and the slurry formulation. The crystallisation
668 kinetics matter little within the ranges considered, as long as the exotherm
669 peak lies above the sintering range. All of these measurements are routine,
670 and most of them can be made on single-material coupons printed with the
671 same slurries.

672 *5.5. Implications for three-dimensional packages*

673 A fired body of this kind has the properties that made glass-ceramic/copper
674 attractive for high-end packaging: a thermal expansion of 3.0 ppm K⁻¹, close
675 to that of silicon (2.6 ppm K⁻¹) [71], a permittivity of about 5 [62], and copper
676 conductors. The model predicts 96% IACS for the conductor from its porosity
677 alone. IBM reported $2.0 \pm 0.2 \mu\Omega$ cm, about 86% IACS, for copper co-fired in
678 H₂/H₂O [62]; the difference points to impurities and grain-boundary oxygen,
679 which the model does not include. Printing adds the geometric freedom
680 that tape lamination lacks. Vertical coils, shielded lines, cavities and ground

681 planes of arbitrary shape can be built into one part, which is relevant to
682 antenna-in-package modules and millimetre-wave front ends [3, 2] and to
683 substrates for wide-bandgap power devices [4]. The present study defines a
684 candidate firing route for such parts and the few quantities that have to be
685 measured before it is tried.

686 6. Conclusions

687 A model of binder removal, sintering and crystallisation was used to
688 design the co-firing of copper inside a cordierite-type glass-ceramic printed as
689 one part by multi-material vat photopolymerisation. It was verified against
690 closed-form solutions, and its glass was calibrated against the published IBM
691 glass-ceramic/copper process, whose atmosphere and burnout temperature it
692 reproduces. No experiments were made. The main conclusions are as follows.

- 693 1. Of the atmospheres examined, only steam removes the carbon from the
694 glass before its pores close. Hydrogen tied to the steam at a fixed fraction
695 of the Cu/Cu₂O equilibrium keeps the copper metallic and reproduces the
696 IBM operating point. At 780 °C such an atmosphere gasifies char seven
697 times faster than humidified forming gas, which leaves more than 300 ppm
698 of carbon in the glass-ceramic in every programme examined.
- 699 2. Fine copper and the glass densify in different temperature ranges: 450–
700 700 °C for 3 µm copper against 850–930 °C for the glass. Co-fired with
701 the IBM-type programme, the copper densifies during the burnout, the
702 free-strain mismatch reaches 15.6%, and a buried line is overloaded in
703 tension. The final shrinkages differ by only 4%; the difficulty lies in the
704 order in which the two materials shrink.

- 705 3. Char holds copper back, so the burnout step also releases the copper to
706 sinter. An optimised paste and programme exploit this coupling: a short,
707 hot burnout just below the coalescence of the glass releases both materials
708 together and reduces the peak mismatch to 3.7%. The coincidence is sharp.
709 A factor of two in the gasification rate, or 15 K in the glass transition,
710 restores most of the mismatch, and the 24% filler it requires limits the
711 copper to 64% IACS.
- 712 4. A design required to meet every constraint for a factor of two either way
713 in the gasification rate, combined with ± 15 K in the glass transition, uses
714 $7\text{ }\mu\text{m}$ copper with almost no filler and a 2.8 h burnout at $806\text{ }^{\circ}\text{C}$. Across
715 that uncertainty it keeps the peak mismatch between 9.3 and 11.4%, the
716 copper at 89% IACS and the carbon at glass closure below 50 ppm, in
717 a 16 h cycle. Robustness costs a factor of two and a half in nominal
718 mismatch.
- 719 5. In three dimensions, a buried copper plane that densifies before the glass
720 acts as a rigid inclusion. The baseline test vehicle shrinks by 15% in plane
721 instead of 23%, and its upper face dishes by 0.6 mm. The nominal optimum
722 reduces the dishing to 0.15 mm and the robust design to 0.32 mm, the
723 same ranking as the one-dimensional mismatch.
- 724 6. On cooling, plastic flow of the copper limits the stress it transmits to its
725 yield stress. A $100\text{ }\mu\text{m}$ copper plane on 1 mm of glass-ceramic bows by
726 $13\text{ }\mu\text{m}$ over 20 mm, about a nineteenth of the elastic prediction.
- 727 7. The inputs that decide the outcome are, in order, the gasification kinetics
728 of the char, the viscosity of the glass, the sintering viscosity of the copper
729 and the char yield. All four can be measured on single-material printed

730 coupons, and they should be measured before the process is tried.

731 **CRedit authorship contribution statement**

732 **[Author names]:** Conceptualization, Methodology, Software, Formal
733 analysis, Visualization, Writing – original draft, Writing – review & editing.

734 **Declaration of competing interest**

735 The authors declare that they have no known competing financial interests
736 or personal relationships that could have appeared to influence the work
737 reported in this paper.

738 **Data availability**

739 The simulation code, input parameters with their provenance, and the
740 scripts that generate every figure in this paper are available from the corre-
741 sponding author on reasonable request.

742 **Declaration of generative AI and AI-assisted technologies in the** 743 **writing process**

744 During the preparation of this work the authors used [name of tool] in
745 order to [reason, e.g. assist with code development, data analysis and drafting
746 of the text]. After using this tool, the authors reviewed and edited the content
747 as needed and take full responsibility for the content of the publication.

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1053 **Appendix A. Copper slab model and its calibration**

1054 The copper conductor uses the slab model developed for single-material
 1055 copper parts printed from a photocurable slurry. Besides the binder chemistry
 1056 of Section 2.2, it tracks the oxygen bound in the powder as Cu_2O . The oxide
 1057 grows in oxygen-bearing gas by a decelerating law fitted to powder oxidation
 1058 data, is reduced by hydrogen above the equilibrium ratio of Eq. (9), and
 1059 reacts with char by $\text{Cu}_2\text{O} + \text{C} \rightarrow 2\text{Cu} + \text{CO}$. None of these oxidation paths is
 1060 active in the co-firing programmes studied here, because the inlet gas contains
 1061 no oxygen and the water-to-hydrogen ratio is held below the copper–oxide

equilibrium. What matters for co-firing is the sintering viscosity. Eq. (7) uses the lattice and grain-boundary diffusion data for copper tabulated by Frost and Ashby [52], with one factor, f_η , fitted to the densification of copper parts printed from a 60 vol% photocurable slurry of 22 μm powder and sintered for 4 h at 1050 $^\circ\text{C}$ in hydrogen, which reached 90–94% relative density [31]. The fit gives $f_\eta = 0.28$, with 0.22–0.34 spanning the reported density range. The powder size enters Eq. (7) through the grain size, set initially to 0.4 times the median particle size, and through the sintering stress. Transferring f_η from 22 μm to the 3–12 μm powders considered here therefore rests on the diffusional scaling of Eq. (7); this is why f_η is included in the sensitivity analysis of Section 4.10.

Appendix B. Verification

Viscous densification of the glass-ceramic. With crystallisation and char removed, and open porosity, Eq. (3) with a Newtonian matrix gives $\dot{\theta} = -9\gamma\theta/(4r\eta)$. The slab model was run through isothermal holds at 810, 830 and 850 $^\circ\text{C}$ with 4, 8 and 16 cells. The simulated porosity follows the closed form, integrated along the simulated part temperature, to within 1.3×10^{-4} over the full range of densification before pore closure, independently of the mesh (Fig. 2a).

Bilayer curvature. The three-dimensional model of a free 12 $\times$ 12 mm plate made of 0.6 mm copper on 1.8 mm glass-ceramic was solved for the instantaneous velocity field at 900 $^\circ\text{C}$, and the curvature rate was fitted to the mid-plane deflection. It converges to the viscous Timoshenko value of Eq. (10)

1085 with second-order behaviour: the error falls from 3.4% at 0.6 mm voxels to
1086 0.69, 0.22 and 0.05% at 0.3, 0.2 and 0.15 mm (Fig. 2b).

1087 *Free sintering and mass conservation.* A free, frictionless cube held isother-
1088 mally densifies homogeneously and isotropically, matches the closed-form
1089 Skorohod–Olevsky solution to 0.22% in density, and conserves solid mass to
1090 0.07%.

1091 *Crystallisation kinetics.* With the rate constant calibrated as described in
1092 Section 2.3, the simulated exotherm at 10 K min^{-1} peaks within 0.01 K of
1093 the prescribed temperature for Avrami exponents of 1, 2 and 3.